Storage Access and Buffer Management

Storage Access

Key Concepts

- **Blocks** are the fundamental units of storage allocation and data transfer.
- Goal: **minimize block transfers** between disk and memory.
- Strategy: keep as many blocks as possible in **main memory**.
- **Buffer:** portion of main memory that stores copies of disk blocks.
- **Buffer Manager:** subsystem that manages buffer allocation and block replacement.

Buffer Manager

- Programs call the buffer manager when they need a block.
- If block is already in buffer → return memory address.
- If block is not in buffer:
  1. Allocate buffer space.
  2. Replace an existing block if needed.
  3. If replaced block was **modified**, write it back to disk.
  4. Read requested block into buffer.
  5. Return its memory address.
- Supports **Buffer Replacement Strategy** (explained below).

Pinned Blocks & Locking

Pinning & Locking Rules

- **Pinned Block:** cannot be written back until unpinned.
- **Pin Count:** block can be evicted only if pin count = 0.
- **Locks:**
  - Shared lock → multiple readers allowed.
  - Exclusive lock → only one updater allowed.
  - Shared and exclusive locks cannot coexist.

Buffer-Replacement Policies

1. Least Recently Used (LRU)

- Idea: use past access pattern to predict future.
- Can be inefficient for queries with sequential scans (e.g., nested-loop join).

2. Toss-Immediate

- Frees a block as soon as its last tuple is processed.

3. Most Recently Used (MRU)

- Keeps block pinned while being processed.
- After processing → block is unpinned and becomes MRU.

4. Heuristic-Based Strategies

- Example: Data dictionary blocks are accessed frequently → keep them in memory.
- Query optimizer can give hints for replacement.

Optimization of Disk Block Access

Techniques

- **Forced Output:** buffer managers may force block writes for recovery.
- **Nonvolatile Write Buffers:** use NVRAM/flash to speed up writes.
- **Write Reordering:** reduce disk arm movement, but must avoid corruption.
- **Log Disk:** sequentially logs block updates (like journaling).
- **Journaling File Systems:** ensure safe reordering by logging updates.

Summary

Concept	Description
Storage Blocks	Basic units of storage allocation and transfer
Buffer	Portion of main memory storing disk block copies
Buffer Manager	Handles allocation, replacement, and consistency
Pinned Block	Block cannot be replaced until unpinned
Locks	Shared (read) vs Exclusive (write) for concurrency
Replacement Policies	LRU, MRU, Toss-immediate, Heuristics
Optimization	Forced output, NVRAM, log disk, journaling